

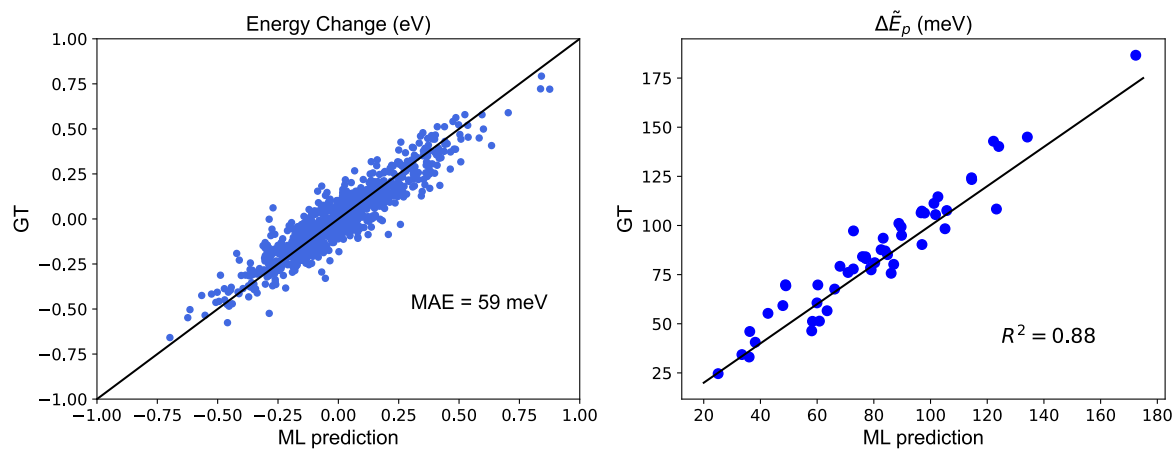


Figure S3: **Evaluation of the GNN model for Peierls barrier prediction on the validation set.**

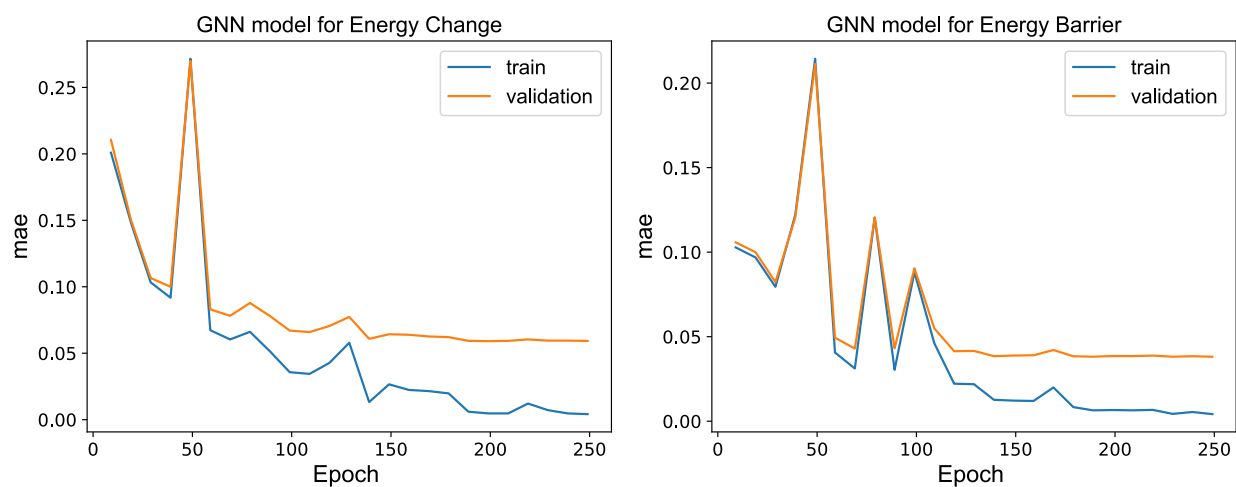


Figure S4: **Loss evolution during training.** Learning curves over 250 epochs for the models trained on (left) potential energy changes and (right) Peierls barrier.